CS & IT ENGINEERING



Algorithms

Heap Algorithms



Lecture No.- 03

By- Aditya sir

Recap of Previous Lecture



- Topic
- Topic

Heap
- deletion
- incr/decr
- Heap Sort

Topics to be Covered



- Topic
- Topic
- Topic

Practice Questions

PYQ



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

Update:

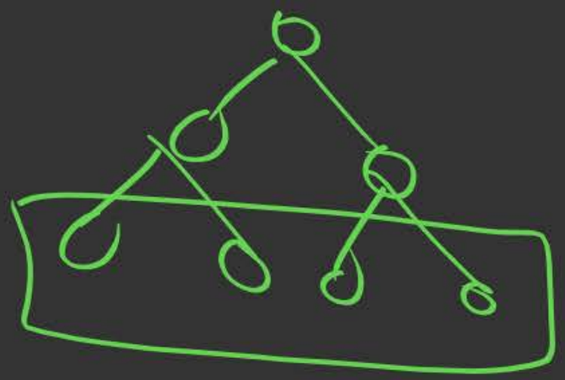
Tomorrow (Sat) : 9 - 11 AM

FLT → after last lec

Imp :- Smallest elem in Max Heap



- $O(n/2) = O(n)$



$\approx \lfloor n/2 \rfloor$ elems

Hw Heapify?



TC: O(n)

Smallest → Max Heap



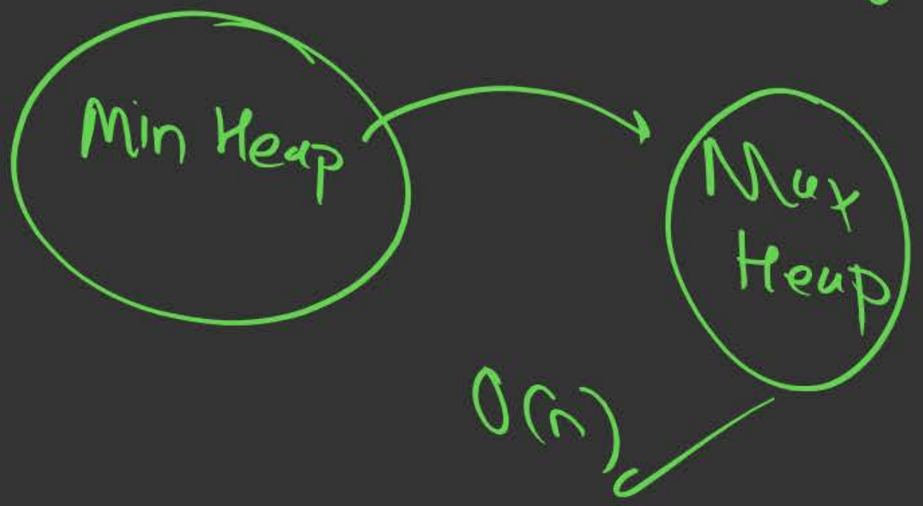
Heapify = O(n)



O(n) + C = O(n)

(Q.2) Largest elem in a Min-Heap?

Same logic :)





Topic : Heap Operations

#Q. Given a Binary Heap with 'n' elements, and it is required to insert 'n' more elements not necessarily one after another into this heap. The total time required for this operation is _____?

A. $O(n^2)$

B. $O(n \log n)$ → 60%

C. $O(n)$ → 26%

D. $O(n^2 \log n)$

Rule : WC Complexity of the Best possible algo (most optimal)

Insertion mtd

Sol.

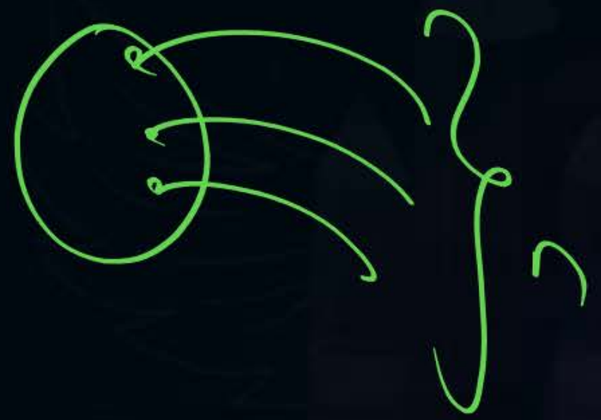


Appr-1.

insertion

n ← 1 → O(logn)

So for n element → O(n * logn)



Appr-2.

Heapify/ Build Heap:

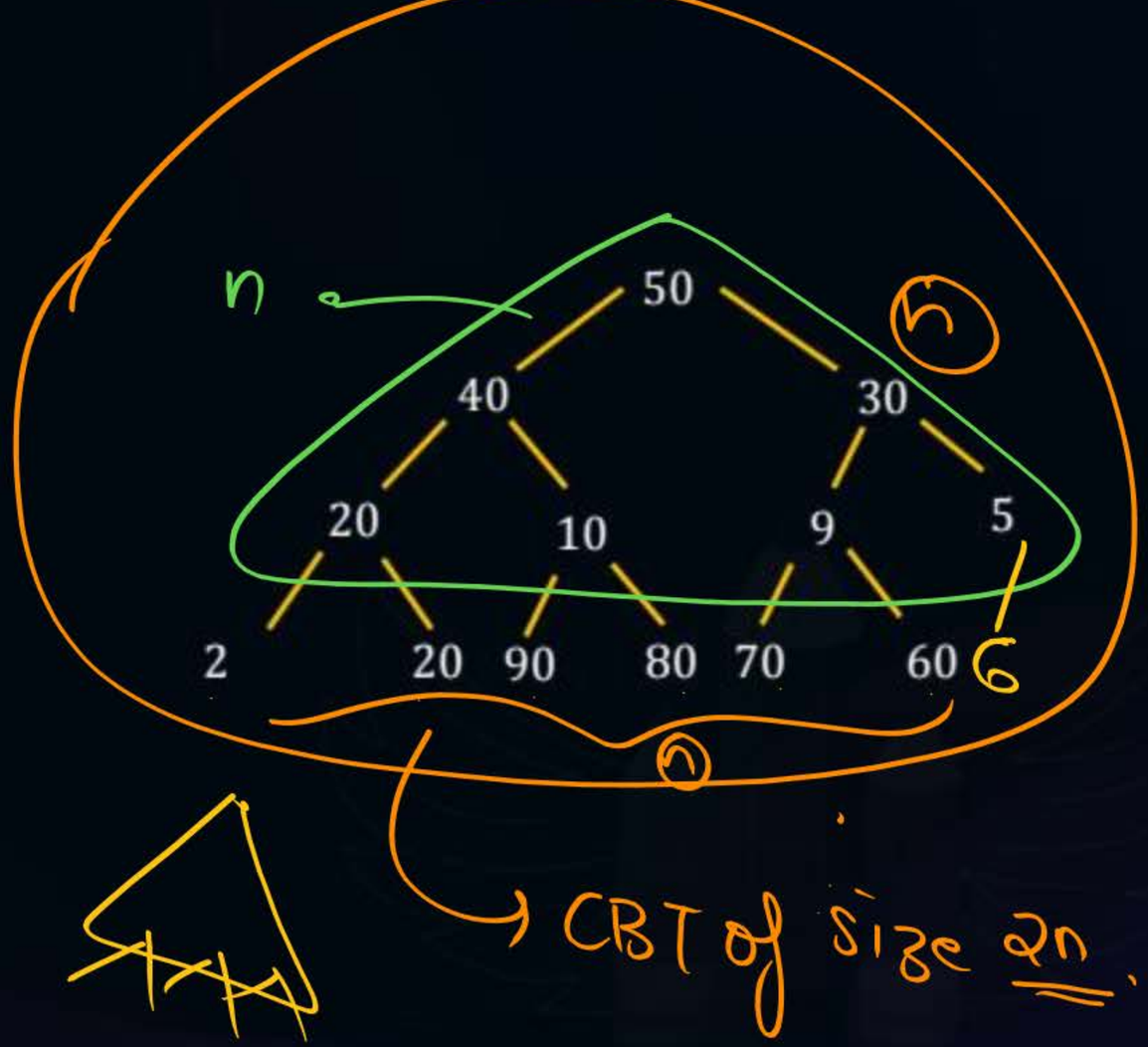
Heap (n)

↓

n new elements

$N \rightarrow O(N)$

$2n \rightarrow O(2n) = O(n)$





Topic : Heap Operations

Min-Heap

#Q. Given binary heap in array with the smallest element at the root, the 7th smallest element can be found in time complexity of _____?

39%

A O(n) ✗

B O(n log n) ✗

C O(log n)

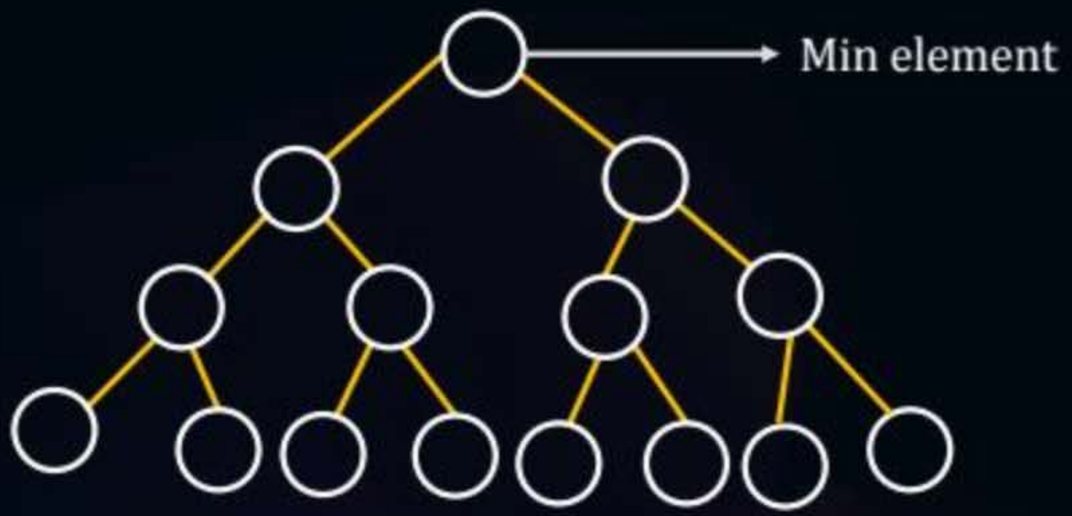
D O(1) ✓

Hint

Appr-1.

Usual / Traditional approach:

Min Heap



1st min → 1 delete → log(n)

2nd min → 1 delete → log(n)

⋮

7th min → 7 delete → log(n)

1 delete operator TC → O(log n)

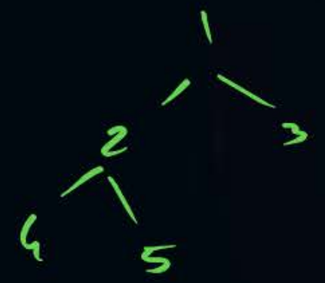
7 delete opr = 7 * log n

= O(log n)

Appr-2.

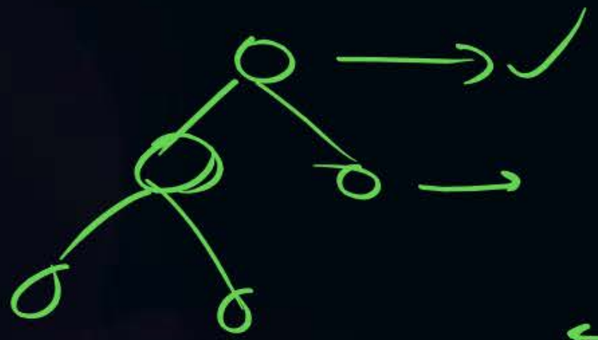
Better / non-traditional

1st min → level 1
2nd min → level 2
3rd min → level 2
→ max depth: level 3









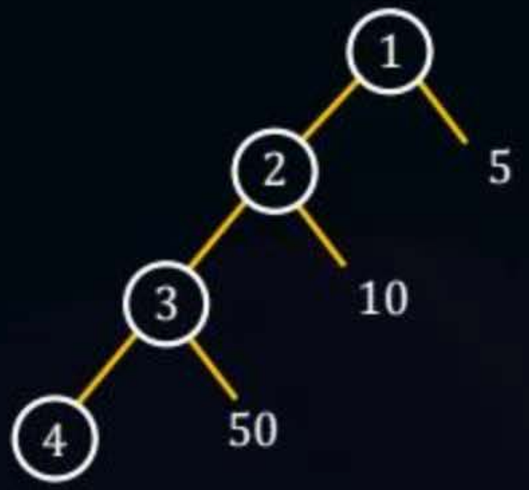


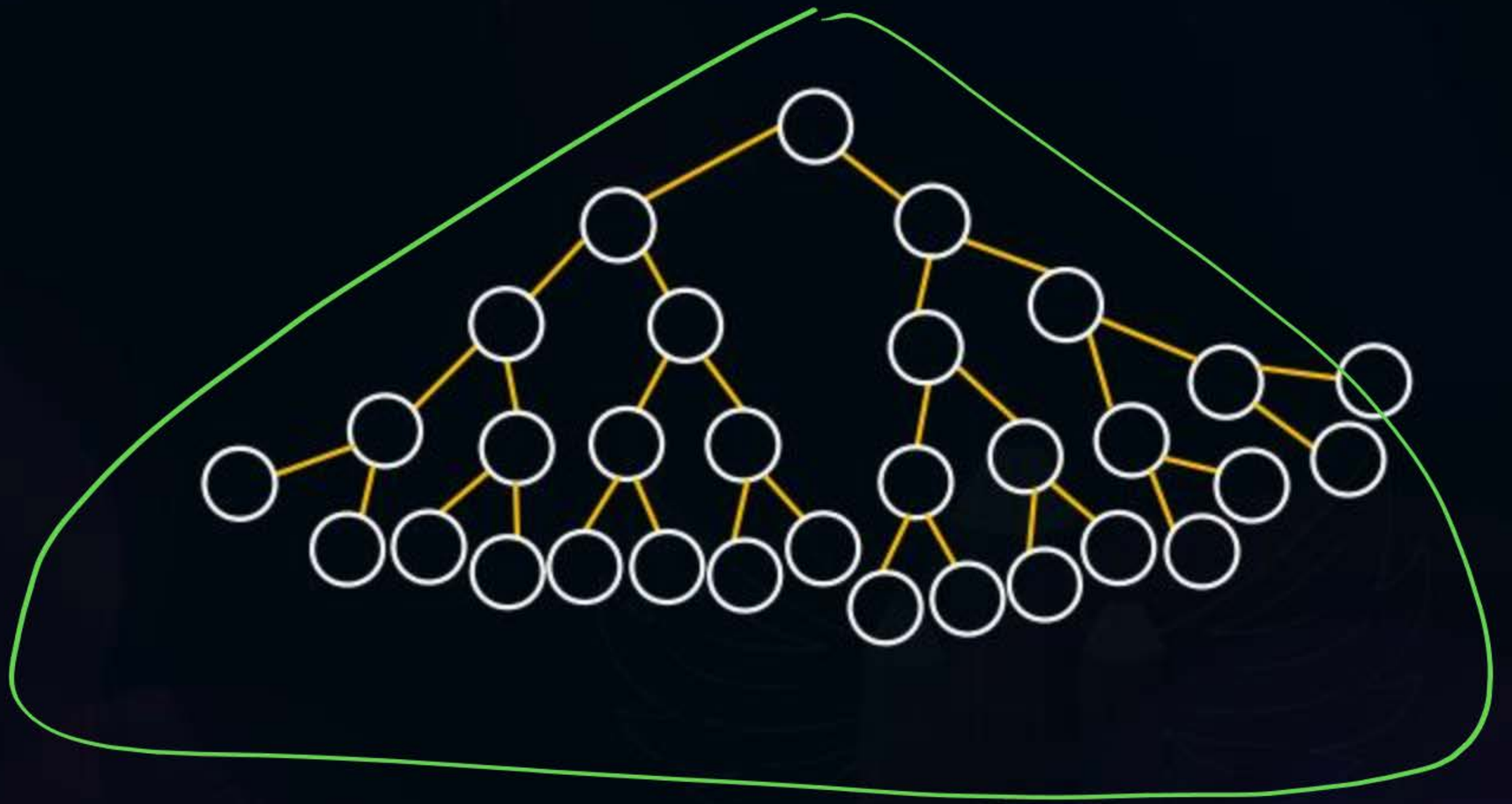
4th min → max depth level 4.

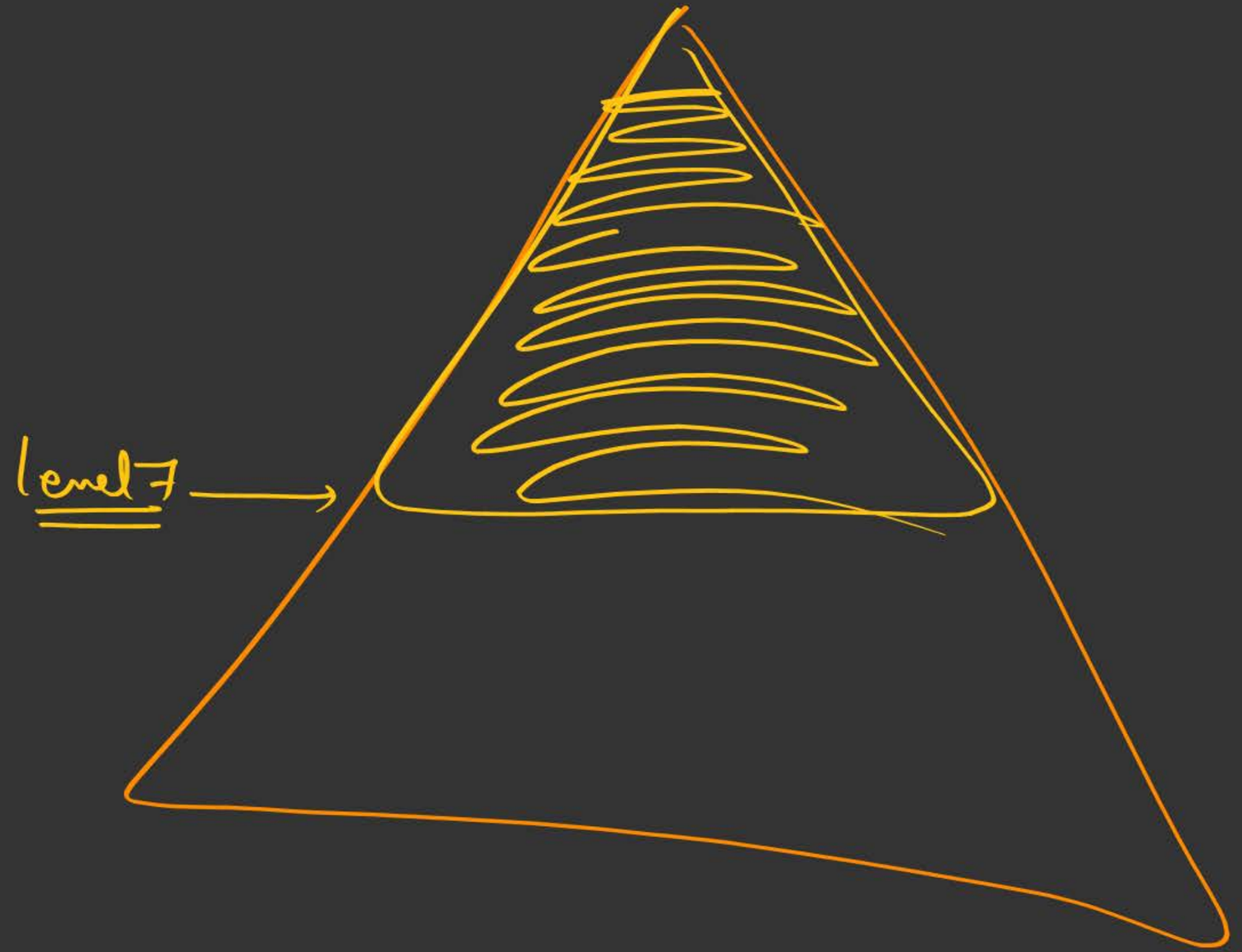
7th min → max depth level 7

Heap → 20 cr elements




1
2
5
3
10
4
50




level 7

We can be sure, that the 7th min element is present within level 1 to level 7.

1st level → max 1 elements → 2^0
2nd level → max 2 elements → 2^1
3rd level → max 4 elements → 2^2

⋮

i^{th} level → max 2^{i-1} elements

L1 → 2^0

L2 → 2^1

L3 → 2^2

L4 → 2^3

L5 → 2^4

L6 → 2^5

L7 → 2^6

So required element present within first 7 levels

$$2^0 + 2^1 + 2^2 \ldots 2^6 = 2^7 - 1$$

Min Heap → n = 200 cr elements

still explore : $2^7 - 1$

'n' → $2^7 - 1$ (const)

→ independent of 'n' = O(1)



Topic : Heap Operations

#Q. Consider binary Heap in an Array with n elements. It is desired to insert an element into the Heap. If a binary search is performed along the path form newly inserted element to the root then the number of comparisons made is order of _____.

33.33%

A O(log n)

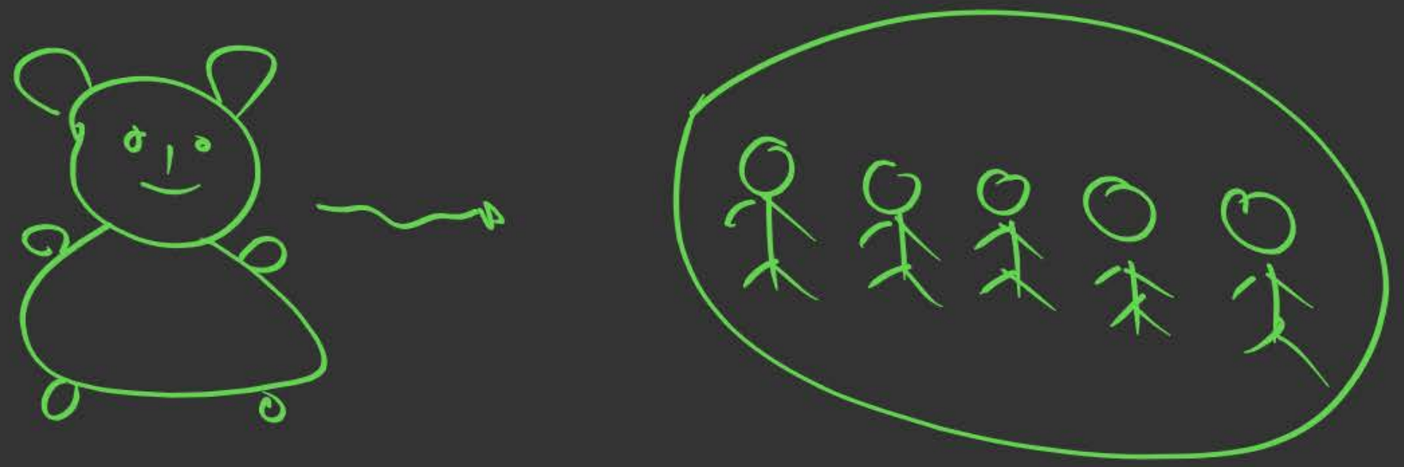
B O(n)

C O(log(log n))

D O(1)

Apti → 15X

JJ → 50




2L
100

Sol.

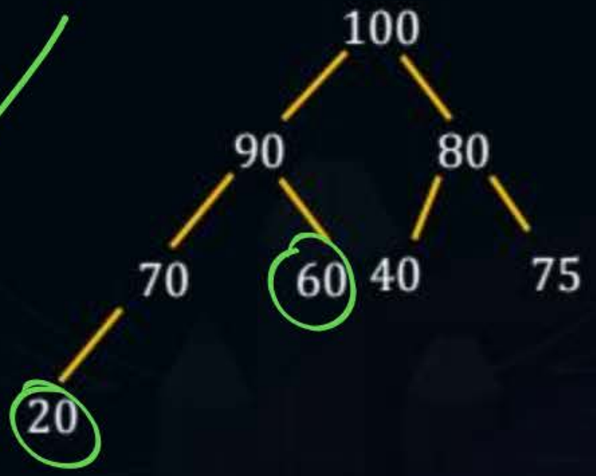
max number of elements in the path from new node to root if Heap has n elements ?

(any)

(path)

n elements → O(log n)

log n elements → O(log (log n))



(log n) Sorted elems

x sorted list $\longrightarrow$ TC : $O(\log(x))$

$x = \log(n)$

TC : $O(\log(\log n))$



Topic : Heap Operations

V.V.V. interesting

#Q. The approximate number of element that can be sorted in O(log n) time using Heap sort is _____.

44%

A $O(n)$ ✗

B $O(1)$ ✗

C $O\left(\frac{\log n}{\log(\log n)}\right)$ ✓

D $O(\log n)$ ✗

TC: O(nlogn)

Topper

Sol.

We know, n element can be sorted in O(n log n) time uasing Heap sort.

'n' elements → O(n log n) time.

Check option D:

no. of elements n → O(n log n)

O(log n) → O(log n * log(log n))

logic.

to sort n element → time O(n log n)

Check option C:

$$x = \frac{\log n}{\log(\log n)}$$

$x \rightarrow n \log n$

$$= \left[\frac{\log n}{\log(\log n)}\right] * \log\left[\frac{\log n}{\log(\log n)}\right]$$

$$= \left(\frac{logn}{\log(logn)}\right)^* [\log(logn) - \log(\log(logn]$$

$$= \log n - \frac{logn * \log\log logn}{\log logn}$$

$= O(\log n)$

$$n - \frac{n \times \log\log n}{\log n} < 1$$



Topic : Heap Operations

H.W

#Q. An operator delete (i) for a binary heap data structure is to be designed to delete the item in the i-th node. Assume that the heap is implement in an array and I refers to the i-th index of the array. If the heap tree has depth d (number of edges on the path from the root to the farthest leaf), then what is the time complexity to re-fix the heap efficiency after the removal of the element?

A O(1)

B O(d) but not O(1)

C $O(2^d)$ but not O(d)

D $O(d2^d)$ but not $O(2^d)$



Topic : Heap Operations

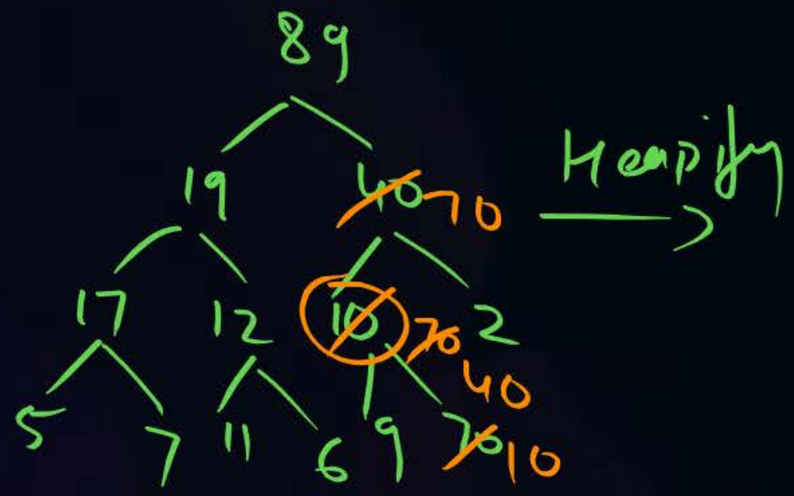
#Q. The minimum number of interchanges needed to convert the array into a max-heap is

89, 19, 40, 17, 12, 10, 2, 5, 7, 11, 6, 9, 70

78%

swaps

- A 0
- B 1
- C 2
- D 3



2

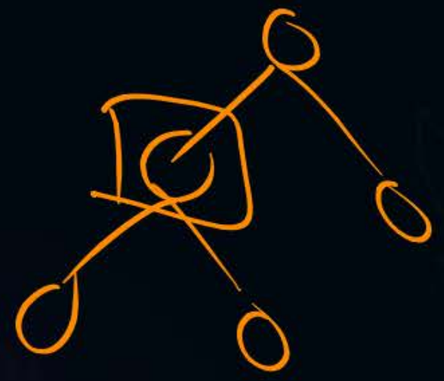


Topic : Heap Operations

#Q. An array of integers of size n can be converted into a heap by adjusting the heaps rooted at each internal node of the complete binary tree starting at the node $\lfloor (n-1)/2 \rfloor$ and doing this adjustment up to the root node (root node is at index 0) in the order $\lfloor (n-1)/2, \rfloor, \lfloor (n-3)/2 \rfloor$ ...., 0. The time required to construct a heap in this manner is

49%

Heapify



A O(log n)

B O(n)

C O(n log log n)

D O(n log n)

Sol.

This is formal explanation process of build Heap / Heapify process.

$n \rightarrow O(n)$



Topic : Heap Operations

#Q. An array X of n distinct integers is interpreted as a complete binary tree. The index of the first element of the array is 0. If only the root node does not satisfy the heap property, the algorithm to convert the complete binary tree into a heap has the best asymptotic time complexity of

62%

A. O(n)

B. O(log n)

C. O(n log n)

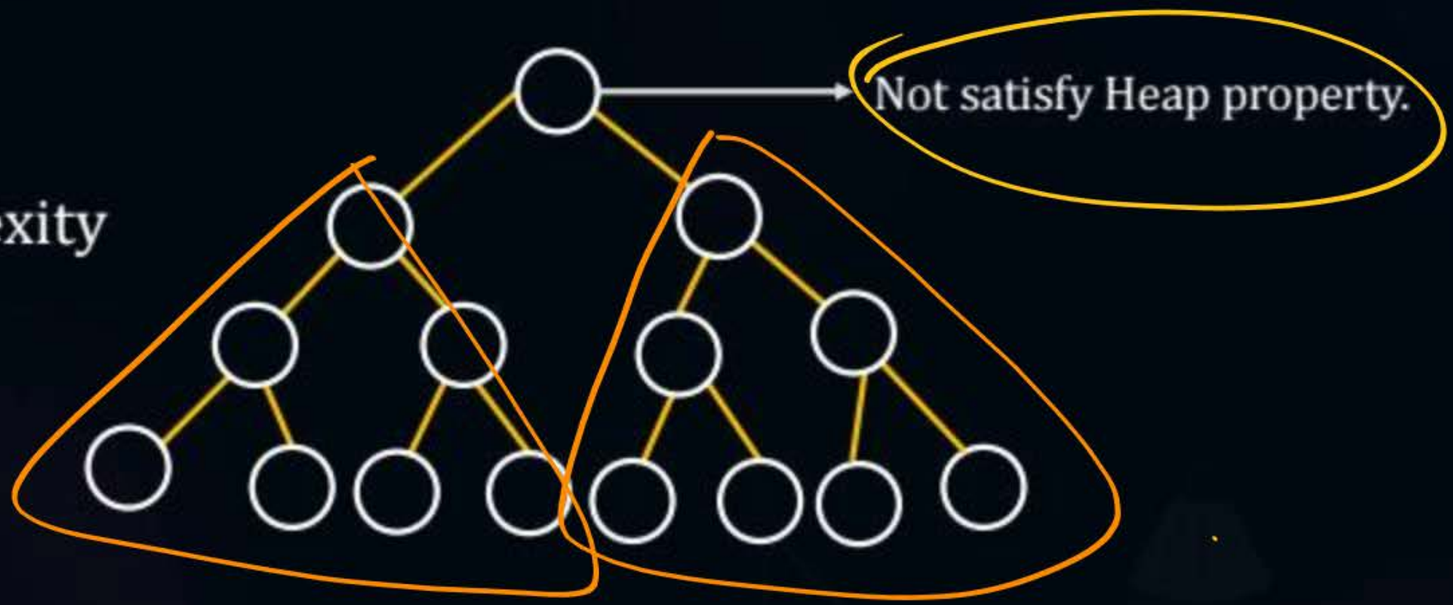
D. O(n log log n)

IIT B

Sol.

best asymptotic complexity

Best $\begin{pmatrix} WC1 \\ WC2 \\ WC3 \end{pmatrix}$



1. Appr 1: Heapify : O(n)

2. Appr 2: adjust root node: → log (n)





Topic : Heap Operations

#Q. Consider a complete binary tree where the left and right subtrees of the root are max-heaps. The lower bound for the number of operations to convert the tree to a heap is

65 %

Ω(1)

A

A. $\Omega(\log n)$

B. $\Omega(n)$

C. $\Omega(n \log n)$

D. $\Omega(n^2)$

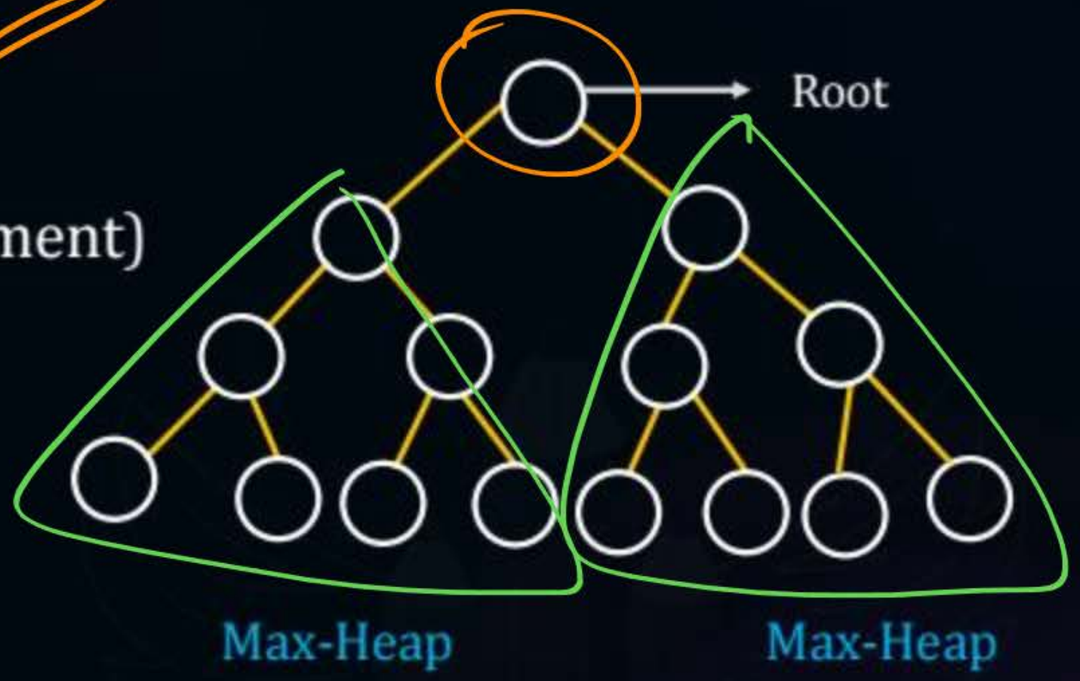
Sol.

Tight lower bound.

$\Omega(1)$

Root → also valid heap (Root → max element)
or Root 2nd largest element → 1 swap.

log n time to adjust



Boot wc
= log(n)





Topic : Heap Operations

HW

#Q. Consider a max heap, represented by the array:
40, 30, 20, 10, 15, 16, 17, 8, 4

Array index	1	2	3	4	5	6	7	8	9
Value	40	30	20	10	15	16	17	8	4

Now consider that a value 35 is inserted into this heap. After insertion, the hew heap is

A 40, 30, 20, 10, 15, 16, 17, 8, 4, 35

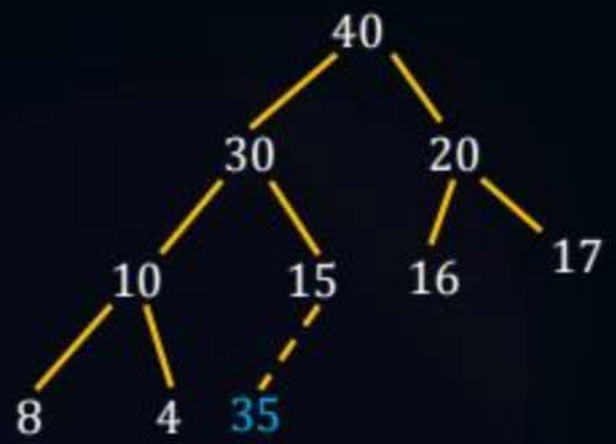
B 40, 35, 20, 10, 30, 16, 17, 8, 4, 15

C 40, 30, 20, 10, 35, 16, 17, 8, 4, 15

D 40, 35, 20, 10, 15, 16, 17, 8, 4, 30

Sol.

40 35 20 10 30 16 17 8 4 15





Topic : Heap Operations

V. interesting

#Q. The number of possible min-heaps containing each value from {1, 2, 3, 4, 5, 6, 7} exactly once is _____.

27.6 %

2018 → 2M

Sol.

n = 7

Appr 1:- Logic → PnC : min Heap Structure

Given

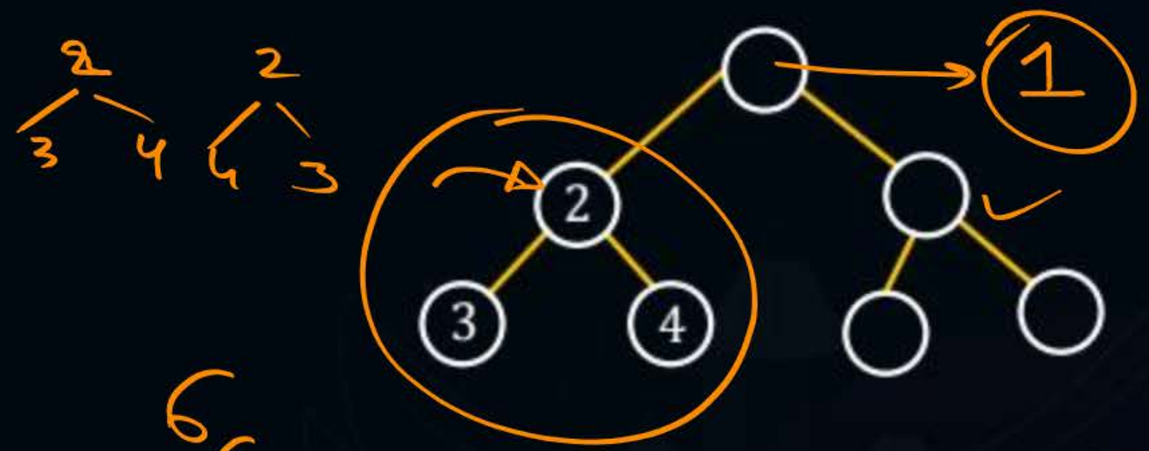
{1, 2, 3, 4, 5, 6, 7}

↓

remaining 6 elements

$^6C_3 * 2! * 2!$

$= 5 \times 4 \times 2 \times 2 = 80$



6C3

Shortest → Algo approach Count of min/max Heap



$$T(n) = {}^{(n-1)}C_k * T(k) * T(n-k-1)$$

where k = no. of nodes in the left subtree

n = total no. of elements

$${}^{n-1}C_k * T(n-k-1) \times T(k)$$

n element → min → at root

remaining (n – 1) element

$${}^{n-1}C_k$$

For prev. e.g:

n = 7, k = 3

$$T(7) = {}^{6}C_3 * T(3) * T(3) = 20 \times 2 \times 2 = 80$$

$T(1) = 1$

$T(3) = 2$

$T(3) = {}^{2}C_1 \times T(1) \times T(1) \times T(1)$ $= 2 \times 1 \times 1$

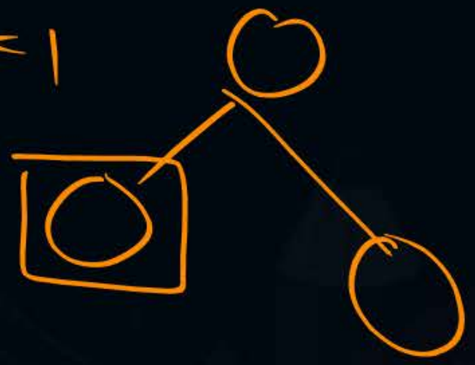
$= 2$



$n = 3$
$k = 1$

$T(1) = 1$

$n = 3$
$k = 1$





2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Heap → PYQ



THANK - YOU